Tutorial 7: Real Trigonometric Integrals

Using a suitable contour integral, evaluate the following trigonometric integrals.

(1) $\displaystyle\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos\theta}\,d\theta$

(2) $\displaystyle\int_0^{2\pi} \frac{4}{5+4\cos\theta}\,d\theta$

(3) $\displaystyle\int_0^{2\pi} \frac{d\theta}{2-\cos\theta}$

(4) $\displaystyle\int_0^{2\pi} \frac{d\theta}{(5-4\cos\theta)^2}$

(5) $\displaystyle\int_0^{2\pi} \frac{\cos 2\theta}{5-3\cos\theta}\,d\theta$

(6) $\displaystyle\int_0^{2\pi} \frac{d\theta}{(1+\frac{1}{2}\sin\theta)^2}$

(7) $\displaystyle\int_0^{2\pi} \frac{\cos\theta}{3+\sin\theta}\,d\theta$

(8) $\displaystyle\int_0^{2\pi} \frac{\cos\theta}{a^2+b^2-2ab\cos\theta}\,d\theta$, $b>a$, $a,\, b\in\mathbb{R}^+$.

Tutorial 8: Cauchy Principal Value

Find the Cauchy principal value of the following integrals if they exist.

(1) $\displaystyle\int_0^{2\pi} \frac{d\theta}{1+2\cos\theta}$

(2) $\displaystyle\int_0^{2\pi} \frac{d\theta}{(1-\cos\theta)^2}$

(3) $\displaystyle\int_0^{2\pi} \frac{\cos 3\theta}{3-5\cos\theta}\,d\theta$

(4) $\displaystyle\int_0^{2\pi} \frac{d\theta}{1-2\cos\theta}$